

Risk-averse restoration of coupled power and water systems with small pumped-hydro storage and stochastic rooftop renewables

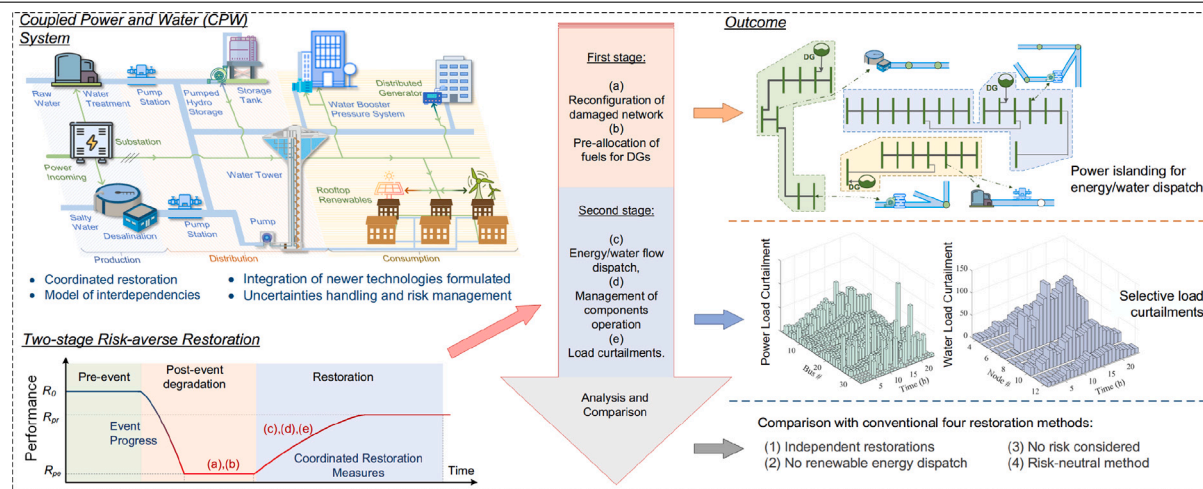
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GRAPHICAL ABSTRACT



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ABSTRACT

Modern coupled power and water (CPW) systems exhibit increasing integration and interdependence, which challenges system performance to disasters and makes service restoration complex during post-disruption. Meanwhile, new technologies, such as small pumped-hydro storage (PHS) and rooftop renewables, are being widely installed and further deepen the interdependencies. To capture these features and improve overall performance, this paper proposes a coordinated restoration framework for a CPW system to respond to disruptions. The proposed CPW model comprises physical networks and mechanisms, considering available units, such as water desalination/treatment plants, pump stations and small PHS, in the water system, and rooftop renewables, distributed generators, in power system. The interdependencies are modeled through component-wise connections and consumer behavior, then grouped into three phases: production, distribution, and consumption. Aggregate service loss with respect to different consumer loads and time periods, is chosen as performance metric and to be minimized using network reconfiguration, energy/water dispatching, load curtailment, and operation management of components. A two-stage risk-averse stochastic programming is applied for reliable restoration and manage risks, to tackle the uncertainties in renewable power generations

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and water/power demands that affect method effectiveness. Finally, the method is implemented on a modified 33-bus/25-node CPW system, and the results demonstrate the effectiveness of the proposed restoration framework.

Nomenclature

Sets and Indices

T, t	Set and index for time intervals
S, s	Set and index for scenarios
$N, i/j$	Set and indices for power buses
$E, (i, j)$	Set and index for power branches
M, m	Set and index for distributed generators
$\mathcal{N}, i/j$	Set and index of water nodes
$\mathcal{E}, (i, j)$	Set and index for water pipelines
$\mathcal{T}, k$	Set and index for water storage tanks
$\mathcal{U}, u$	Set and index for pumps
$\mathcal{O}, o/\mathcal{O}', o'$	Set and index for water desalination and treatment plants

Parameters

ΔT	Time interval
ρ, g	Water density and acceleration due to gravity
η_i^p, η_i^t	PHS pump and turbine efficiency at node i
$\mathcal{H}_{r,i}^t$	Head rise required by water tank at node i
r_{ij}/x_{ij}	Resistance/reactance of branch (i, j)
$S_{ij}^{\max}$	Capacity of branch (i, j)
$P_i^{t,s}/Q_i^{t,s}$	Active/reactive load at bus i
$V^{\max}/V^{\min}/V_0$	Maximum/minimum/reference voltage
$P_{DG,m}^{\max}/Q_{DG,m}^{\max}$	Maximum power for DG m , MW/MVar
$E_{DG,m}^{\max}$	Maximum fuel available for DG m , MWh
$E_{ex}^{\text{cap}}/\Xi_{m,t}$	Capacity of stock fuel and arrival time to DG m
$\varphi_i/\varphi_{w,i}/\varphi_{pv,i}$	Power factor of load and wind/solar power
$\mathcal{W}_i^{t,s}$	Water load at node i in time t , m^3
$\mathcal{Q}_{R,o}^{\max}/\mathcal{Q}_{R,o}^{\min}$	Maximum/minimum inflow of water treatment plant o , m^3/h
$\rho_o^{\max}$	Maximum water conversion ratio of desalination plant o
$r_{o'}^f$	coefficient for energy consumption of water treatment, m^3/MWh
M^P	Big scatter for power side model
G	Coefficient of Hazen–Williams equation
$\mathcal{V}_k^0/\mathcal{V}_k^{\max}$	Initial/maximum volume for tank k , m^3
$\mathcal{E}\mathcal{L}_i$	Elevation of node i , m
$\mathcal{P}_i^{\text{des}}/\mathcal{P}_i^{\min}$	Design/minimum pressure of node i , m
$\ell_{o,i}$	Leakage coefficient of node i , $m^3/(s \cdot m)$

γ_i^p/γ_i^c	Pressure-driven/leakage exponent of node i
$f_{u,a}/f_{u,b}/f_{u,c}$	Coefficients for pump equations
$\Gamma_u^{\max}$	Maximum power for pump u
$P_{P,i}^{\max}/P_{P,i}^{\min}$	Boundaries of power consumption by small PHS at node i
$P_{\mathcal{H},i}^{\max}/P_{\mathcal{H},i}^{\min}$	Boundaries of power generation by small PHS at node i
$f_{o,a}/f_{o,b}$	Coefficients for water desalination equations
$\varpi_{i,i}^{N,N}/\xi_{i,i}^{N,N}$	Coefficients for power/water consumption nexus
c_i	Binary acceptance level for delayed demand
$P_{w,i}^{t,s}/P_{pv,i}^{t,s}$	Rooftop wind/solar power generation capacity
$\omega_{w,i}^t/\omega_{p,i}^t$	Weight of water/power loads at i/i
α/λ	Parameters in Conditional Value at Risk (CVaR) formulation
$N_{rt,i}/N_{ft,i}$	Binary status for potential root bus

Variables

α_{ij}^p	Binary state for branch (i, j) , 1 if connected
β_i^p	Binary state for bus i , 1 if root bus
F_{ij}	Fictitious flow from i to j
$E_{ex,m}$	Non-negative fuel pre-allocation for DG m
$P_{r,i}^{t,s}/Q_{r,i}^{t,s}$	Power load restored at bus i
$P_{ij}^{t,s}/Q_{ij}^{t,s}$	Power flow from bus i to bus j
V_i^t	Voltage magnitude at bus i
$P_{DG,m}^{t,s}/Q_{DG,m}^{t,s}$	Dispatched power of DG m
$E_{DG,m}^{t,s}$	Remaining fuel of DG m at time t
$\mathcal{W}_{r,i}^{t,s}/\mathcal{W}_{l,i}^{t,s}$	Restored water load and water leakage at node i
$\mathcal{H}_i^{t,s}/\mathcal{P}_i^{t,s}$	Water head/pressure at node i
$\mathcal{Q}_{i,j}^{t,s}$	Water flow from node i to j
$\mathcal{V}_k^{t,s}$	Stored water at tank k
$\mathcal{C}_k^{t,s}/\mathcal{D}_k^{t,s}$	Water inflow/outflow of tank k
$\mathcal{Q}_{R,o}^{t,s}/\mathcal{Q}_{C,o}^{t,s}/\Gamma_o^{t,s}$	Water inflow/outflow and power consumption of desalination plant o
$\mathcal{Q}_{f,o'}^{t,s}/\Gamma_{o'}^{t,s}$	Water production/power consumption of treatment plant o'
$P_{P,i}^{t,s}/P_{\mathcal{H},i}^{t,s}$	Power consumption and generation from small PHS charging/discharging
$\mathcal{H}_u^{t,s}/\Gamma_u^{t,s}$	Head gain/power consumption of pump u
$\Gamma_{wb,i}^{t,s}$	Power for WBPS at node i
$\Delta UD_i^{t,s}$	Unmet water demand at node i
$Buf_i^{t,s}$	Buffer to record unmet demand of node i
$P_{w,i}^{t,s}/P_{pv,i}^{t,s}$	Rooftop wind/solar power injection
$Q_{w,i}^{t,s}/Q_{pv,i}^{t,s}$	Reactive power of rooftop wind/solar
$Cost_s/\Delta_s/v$	Variables in CVaR formulation

1. Introduction

1.1. Background and literature review

Due to the growing utilization of electricity-driven water utility devices, the water distribution system (WDS) and power distribution system (PDS) are becoming increasingly intertwined [1]. The increased

interdependencies offer the potential for the enhancement of coupled power and water (CPW) system's performance [2]. However, it also renders CPW system more susceptible to consequences of natural disasters and human-caused failures. In particular, for high-impact

low-probability events, e.g., an earthquake or a hurricane, the consequences can be cascading and disastrous. For example, the July 2022 thunderstorms in Paris, Arkansas, U.S., caused power shortages at the water treatment plant and resultant water supply interruptions [3]. In the summer of 2022, the prolonged heat wave in China decreased hydroelectric power generation and some Chinese cities had to ration power [4]. These instances demonstrate the urgent need to increase the coupled WDS and PDS resilience, especially in light of the increasing frequency and severity of triggering events [5]. Since their prolonged interruption can pose a life-threatening risk to vulnerable people, as was the case in Texas during the 2021 winter storm, a robust connection between electricity and water distribution networks is particularly important for the mitigation of such disasters [6].

Resilience can be defined as the capacity of a system to absorb disturbance and reorganize to retain essentially its structures and functions [7]. The concept of restoration is an integral part of resilience, where the goal is to regain infrastructure system's service to a desired level after disruption. In this work, the restoration is considered as an emergency response issue, with the aims to minimize the consequence of disruption and reduce system function degradation using available resources, comprising distributed generation and water tank storage.

Despite the fact that the vulnerabilities of the water and electricity infrastructures are widely recognized, efforts to improve their resilience are mainly focused on methods that regard PDS and WDS separately, without considering their interdependent functioning. For resilience of standalone WDS, common strategies comprise components upgrade and enhancement, increased redundancy, and adaptive operation for WDS management, etc. For example, Ref. [8] described a discrete simulation model for water supply system restoration, considering damage inspection, pipeline rerouting, isolating, and repairing, as applied to a simplified physical model of Los Angeles water supply system. Ref. [9] proposed a Bayes network-based framework to model post-earthquake functional recovery of the WDS network, and evaluate the efficacy of restoration with respect to pipeline connectivity. In Ref. [10], the results of a competition on post-disaster water network restoration were summarized based on an existing water network termed as B-City WDS comprising pipes, pumps and a reservoir. Among the solution methods, three types of methods were commonly applied: meta-heuristic, greedy algorithms, and ranking-based prioritization. However, the results of these methods highly depend on initial solution assumed and frequently suffer from severe convergence issues. For the same WDS, researchers in Ref. [11] further proposed a dynamic optimization framework to identify optimal sequencing of recovery actions. In Ref. [12], a multi-objective programming was proposed on the optimal rerouting of water supply balancing the valve switching number and restored load based on small benchmark systems. These studies implicitly assumed adequate power supply for water system components, and thus decoupled the interdependence between electricity and water infrastructure. However, such disruptions would also affect PDS. The damage and restoration of power system almost dominate water equipment that consumes power, and impact WDS operation [1].

Similar to WDS, the restoration methods for standalone PDS applied on modified IEEE benchmark systems consist of network reconfiguration, mobile energy dispatching, microgrid forming and self-healing operations. The main target is to reduce the load curtailment, without considering the cascading outage in other facilities. For example, Ref. [13] proposed a sequential service restoration framework for PDS suffering from large-scale power outages, where switches, distributed generators, microgrid forming and unbalanced three-phase operations were considered. Ref. [14] proposed a two-stage stochastic program to optimize the resource allocation and crew dispatching for PDS restoration. Ref. [15] took into account demand response and hourly distribution network reconfiguration to minimize the total service loss. Ref. [16] analyzed the potential of rooftop solar energy and battery storage on improving system resilience under heavy and moderate

damages and proposed an optimal dispatching method. In Ref. [17], a data-driven restoration approach was presented considering distributed energy resource (DER) uncertainties, microgrid forming, and load prioritization. Researchers in Ref. [18] proposed a two-stage co-optimization method to combine the repair crew scheduling and dynamic network reconfiguration to minimize the total service loss.

The two previous paragraphs reviewed works on standalone water systems [8–12] and power systems [13–18] with the interdependencies between the two systems neglected. With the increasingly coupled WDS and PDS, it becomes crucial to consider them as a whole CPW system. These interdependencies arise from both component-level demands [1] and consumer side constraints [19]. Refs. [20–24] explored modeling and restoration strategies for CPW systems with interdependencies modeled at varying degree of detail, e.g., physical, topological and with different degrees of connectivity. Ref. [20] discussed a four-stage risk-informed decision-making framework, containing problem definition, condition assessment, alternative comparing and decision analysis, and was applied in the risk mitigation problem of practical CPW system comprising component level interdependencies. Ref. [21] developed a multi-objective program to study the restoration for CPW systems, where the interdependencies are topologically defined instead of physical flows being modeled. Based on such coupled network flow model, researchers in Ref. [22] applied a two-stage stochastic program to address the preparedness planning problem to reduce the vulnerability of the CPW system. In Ref. [23], a simulation–optimization framework for CPW system restoration is developed, where the physical models for PDS and WDS are simulated and optimized separately. Their interdependencies are represented as mutual demands between the two systems. Ref. [24] proposed a mixed integer second order cone programming (MISOCP) based restoration method for CPW systems, with both system components and interdependencies incorporated. However, such works by necessity model the coupling to varying details and neglect system uncertainties. To the authors' knowledge, a trackable CPW model with key physical features incorporating component level interdependencies and system uncertainty is still lacking.

The growing penetration of rooftop renewables and power-related water equipment (small PHS and water booster pressure system (WBPS), etc.) further introduces coupling features and uncertainties in renewable generation [25,26], which requires a reassessment of current restoration approaches. Despite the existing methodologies, there are still relatively fewer reported works for the coordinated restoration of WDS and PDS and the utilization of potential resources and flexibility from such installed devices to improve total restoration. In addition to modeling coupling features also becomes a key challenge to properly address the uncertainties and manage risk. The uncertainties arise from the capacity of renewable generations, and varying demands of power and water loads. The risk refers to the possibility of suffering from unfavorable high costs. Stochastic programming (SP) is a developed method to support decision-making under parametric uncertainties [27]. However, the original SP takes mean cost as the only metric and focuses on average performance over all possibilities [14], regardless of variance among scenarios, which leads to the risk of high cost under some outliers. In this regard, the risk-averse method is adopted here to add risk measures as additional metrics. Hence, this methodology tends to avoid excessively high costs under outlier scenarios [28]. In addition, despite the existing methodologies, there are still relatively fewer reported works for the coordinated restoration of WDS and PDS and the utilization of potential resources and flexibility from installed devices to improve total restoration.

1.2. Contributions

To address the research gap discussed above and improve the overall performance of CPW systems, this paper proposes a coordinated restoration framework for CPW systems with physical mechanisms modeled, critical interdependencies formulated, and new devices considered. The contributions of the paper are summarized as follows:

- The interdependencies between WDS and PDS are investigated and modeled in three phases: production, distribution, and consumption at the component level. At each phase, the couplings are formulated as supply–demand connections between the respective components of the two systems. The couplings of power and water consumption including from renewables are further modeled according to different types of consumer behavior.
- A CPW model is developed based on physical (power and water) mechanisms and interdependencies, taking into account network topology, component features, and varying demands. The proposed method also integrates emerging technologies, comprising small PHS and rooftop solar panels and wind turbines, and analyzes their impacts on the overall resilience of the CPW system.
- A restoration model considering uncertainties is demonstrated to take advantage of extra flexibility and tackle the uncertainties introduced by rooftop renewables and power/water demands.

The reminder of the work is organized as follows: Section 2 presents an overview of the proposed method and describes the interdependencies within CPW system. Risk-averse CPW restoration is formulated in Section 3. The model linearization, scenario generation and reduction, as well as solution method are presented in Section 4. The results of the proposed model on a 33-bus/25-node test system are discussed in Section 5, followed by conclusions in Section 6.

2. Interdependencies formulation

2.1. Overview

In this work, the interdependencies between PDS and WDS are categorized into three phases according to different operation stages: production, distribution, and consumption, as illustrated in Fig. 1.

- At production phase, electricity from power system supports water treatment and desalination plants to produce potable water [29]. These interdependencies connect energy dispatch and overall capacity of water supply for CPW system.
- At distribution phase, produced water is fed into pipe networks. Electricity-driven pump stations work to maintain proper pressure head for whole water distribution network. The water tower needs a booster pump to lift water for restoring and maintaining pressure. For tanks equipped with small PHS device, the interdependencies show in the form of power withdrawn to lift water and power generated from release of water [30]. These components provide energy/water storage and water pressure management. The interdependent flows connect buses in PDS, nodes/links in WDS and provide flexibility to services dispatch.
- Finally, at consumption phase, the water in pipes need to be secondary pressurized to provide water service with proper pressures and volumes for some high-rise buildings [31]. The consumption phase interdependencies are also expressed in consumer behavior, with activities that simultaneously consume both supplies [19]. These interdependencies quantitatively represent intertwined power/water consumption, and further impact the volume of effective service on the consumer side.

Mechanisms for each phase are formulated in following paragraphs. For the ease of explanation, we define a set of incidence matrices, Λ , to represent the linkage of components of the WDS and PDS, for example, $\Lambda_{i,j}^{N,N} = 1$ denotes that power bus i is connected to water node j , and so on for other connections.

2.2. Production phase:

2.2.1. Desalination plant

Water desalination consumes electricity to remove salts from seawater to converting it into fresh water [32]. Here, the reverse osmosis

type of desalination is considered. The power consumption for the desalination unit is indicated in (1) to (2) [32].

$$\Gamma_{\phi}^{t,s} = \left[f_{\phi,a} \frac{2Q_{R,\phi}^{t,s} - Q_{C,\phi}^{t,s}}{Q_{R,\phi}^{t,s} - Q_{C,\phi}^{t,s}} + f_{\phi,b} Q_{R,\phi}^{t,s} \right] Q_{C,\phi}^{t,s}, \quad \forall \phi \in \mathcal{O}, t \in \mathcal{T}, s \in \mathcal{S} \quad (1)$$

$$\begin{cases} Q_{R,\phi}^{\min} \leq Q_{R,\phi}^{t,s} \leq Q_{R,\phi}^{\max}, & \forall \phi \in \mathcal{O}, t \in \mathcal{T}, s \in \mathcal{S} \\ 0 \leq Q_{C,\phi}^{t,s} \leq \rho_{\phi}^{\max} Q_{R,\phi}^{t,s}, & \forall \phi \in \mathcal{O}, t \in \mathcal{T}, s \in \mathcal{S} \end{cases} \quad (2)$$

2.2.2. Water treatment

Water treatment plants process the raw water into potable water, which is modeled as a power consumer and freshwater producer [29]. Herein, the energy consumption for water treatment is linearized in (3).

$$\Gamma_{\phi'}^{t,s} = r_{\phi'}^f Q_{f,\phi'}^{t,s}, \quad \forall \phi' \in \mathcal{O}', t \in \mathcal{T}, s \in \mathcal{S} \quad (3)$$

2.3. Distribution phase:

2.3.1. Small pumped-hydro storage

A PHS unit serves as a link between the PDS and the WDS, interacting with both systems to maximize the benefits of their interdependence. A small PHS unit with controllability has the potential to provide additional flexibility to both systems [30].

Here, the PHS unit is modeled and integrated into CPW system as indicated in Fig. 1. It consists of a water storage tank linked to one of the WDS nodes for water exchange, and a pressure booster pump to lift water from distribution network to the tank. Stored water can be released back into the WDS depending on the water supply and power generation requirements. The discharged water powers the PHS unit turbine, which produces electricity.

The power consumption and generation of small PHS are modeled in (4) to (8). Eq. (9) indicates tank volume boundaries and (10) constrain the age for stored water, where the residence time should not exceed T for water quality consideration.

$$\eta_i^P P_{P,i}^{t,s} = \rho g (\mathcal{H}_{r,i} - \mathcal{H}_i^{t,s}) Q_{\mathcal{H},i}^{t,s} \Lambda_{\mathcal{H},i}^{\mathcal{T},N}, \quad \forall i \in \mathcal{N}, t \in \mathcal{T}, s \in \mathcal{S} \quad (4)$$

$$P_{\mathcal{H},i}^{t,s} = \eta_i^{\mathcal{H}} \rho g (\mathcal{H}_{r,i} - \mathcal{H}_i^{t,s}) Q_{\mathcal{H},i}^{t,s} \Lambda_{\mathcal{H},i}^{\mathcal{T},N}, \quad \forall i \in \mathcal{N}, t \in \mathcal{T}, s \in \mathcal{S} \quad (5)$$

$$\mathcal{V}_{\mathcal{H}}^{t,s} = \mathcal{V}_{\mathcal{H}}^{t-1,s} + (\mathcal{Q}_{\mathcal{H}}^{t,s} - \mathcal{Q}_{\mathcal{H}}^{t,s}) \Delta T, \quad \forall \mathcal{H} \in \mathcal{T}, t \in \mathcal{T}, s \in \mathcal{S} \quad (6)$$

$$P_{P,i}^{\min} \leq P_{P,i}^{t,s} \leq P_{P,i}^{\max}, \quad \forall i \in \mathcal{N}, t \in \mathcal{T}, s \in \mathcal{S} \quad (7)$$

$$P_{\mathcal{H},i}^{\min} \leq P_{\mathcal{H},i}^{t,s} \leq P_{\mathcal{H},i}^{\max}, \quad \forall i \in \mathcal{N}, t \in \mathcal{T}, s \in \mathcal{S} \quad (8)$$

$$0 \leq \mathcal{V}_{\mathcal{H}}^{t,s} \leq \mathcal{V}_{\mathcal{H}}^{\max}, \quad \forall \mathcal{H} \in \mathcal{T}, t \in \mathcal{T}, s \in \mathcal{S} \quad (9)$$

$$\mathcal{V}_{\mathcal{H}}^0 \leq \sum_{t \in \mathcal{T}} \mathcal{Q}_{\mathcal{H}}^{t,s} \Delta T, \quad \forall \mathcal{H} \in \mathcal{T}, s \in \mathcal{S} \quad (10)$$

2.3.2. Pump station

The pump station model contains two equations for head gain and coordinate power consumption [33], as in (11) to (13).

$$\mathcal{H}_u^{t,s} = f_{u,a} \sum_{(i,j) \in \mathcal{E}} Q_{i,j}^{t,s} \Lambda_{u,i,j}^{\mathcal{U},\mathcal{E}} + f_{u,b}, \quad \forall u \in \mathcal{U}, t \in \mathcal{T}, s \in \mathcal{S} \quad (11)$$

$$\eta_u \Gamma_u^{t,s} = f_{u,c} \sum_{(i,j) \in \mathcal{E}} \left(f_{u,a} Q_{i,j}^{t,s 2} + f_{u,b} Q_{i,j}^{t,s} \right) \Lambda_{u,i,j}^{\mathcal{U},\mathcal{E}}, \quad \forall u \in \mathcal{U}, t \in \mathcal{T}, s \in \mathcal{S} \quad (12)$$

$$0 \leq \Gamma_u^{t,s} \leq \Gamma_u^{\max}, \quad \forall u \in \mathcal{U}, t \in \mathcal{T}, s \in \mathcal{S} \quad (13)$$

2.3.3. Water tower

Water towers are storage components in WDS with fixed capacities [34]. A water tower can maintain proper pressure and volume for the water usage of communities. The head and volume change with charging/discharging are formulated in (6) to (10) (same as small PHS).

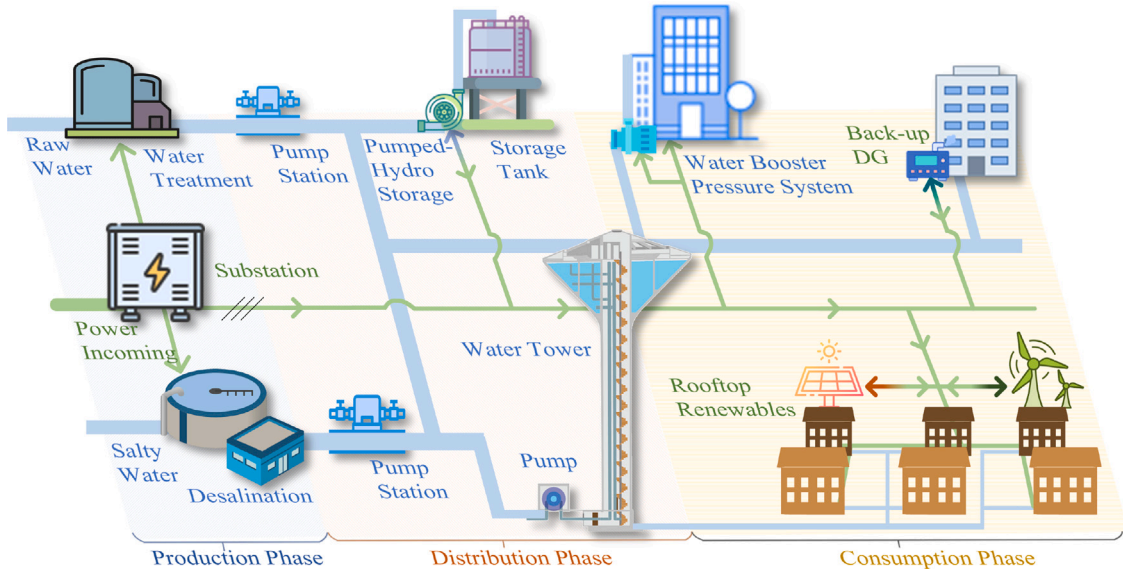


Fig. 1. Conceptual illustration of interdependencies within urban CPW system.

2.4. Consumption phase:

2.4.1. Water booster pressure system

Tall buildings present important challenges in water supply. The operation of WBPS is to pressurize water from WDS and maintain proper pressure and volume for upper floors, which is directly related to water consumption [31] of the building. A dynamic model for WBPS operation is described in (14) to (16).

$$\dot{V}_{f,i}^{t,s} = Q_{in,i}^{t,s} - Q_{out,i}^{t,s}, \quad \forall i \in \mathcal{N}, t \in T, s \in S \quad (14)$$

$$p_{air,i}^{t,s} = \frac{(p_{pr} + p_a) \dot{V}_T}{\dot{V}_T - \dot{V}_{f,i}^{t,s}} - p_a, \quad \forall i \in \mathcal{N}, t \in T, s \in S \quad (15)$$

$$\Gamma_{wb,i}^{t,s} = \frac{1}{c_u \eta_i} \left(p_{air,i}^{t,s} - p_a \right) Q_{in,i}^{t,s}, \quad \forall i \in \mathcal{N}, t \in T, s \in S \quad (16)$$

where, $\dot{V}_T / \dot{V}_{f,i}^{t,s} / \dot{V}_{f,i}^{t,s}$ are the capacity, water volume and dynamic change of the pressure tank; $Q_{in,i}^{t,s} / Q_{out,i}^{t,s}$ are water inflow/outflow of the pressure tank. $p_{pr} / p_{air,i}^{t,s} / p_a$ represent preload pressure of tank, as well as pressure of tank air and atmosphere. c_u and η_i indicate scaling factor and efficiency. More details are in Ref. [31].

In this paper, the dynamic WBPS model is converted to static one based on the assumption that in a macro timescale (hourly), the input and output water flow of pressure tank are balanced, which leads to: $Q_{in,i}^{t,s} = Q_{out,i}^{t,s} = \dot{V}_{f,i}^{t,s}$, and then $\dot{V}_{f,i}^{t,s} = 0$, and $\dot{V}_{f,i}^{t,s}$ becomes a constant value: $\dot{V}_{f,i}^{t,s}$. Then power consumption of booster pump is linear with water supply:

$$\Gamma_{wb,i}^{t,s} = \frac{1}{c_u \eta_i} \left[(p_{pr} + p_a) \frac{\dot{V}_T}{\dot{V}_T - \dot{V}_{f,i}^{t,s}} - 2p_a \right] \dot{V}_{f,i}^{t,s}, \quad \forall i \in \mathcal{N}, t \in T, s \in S \quad (17)$$

2.4.2. Power & water nexus loads

The nexus between power and water consumption is observed both in industry and residential activities. It represents a simultaneous demand for two commodities supplies [19]. Shortage of any one supply can hinder the activity, which in turn limits the consumption of the other. Herein, this feature is modeled as a set of mutual constraints for both power and water restorations. The constraint from power to water restoration is formulated in (18), and similar constraint from water to power is in (19). It is noted that the values of coefficient $\varpi_{i,i}^{N,N}$ and $\xi_{i,i}^{N,N}$ represent both direction and degree of the interdependency. If $\varpi_{i,i}^{N,N} \times \xi_{i,i}^{N,N} = 1$, they are strictly proportional, while when $\varpi_{i,i}^{N,N} \times$

$\xi_{i,i}^{N,N} \geq 1$, the interdependency of two sides becomes uneven. That means the consumption of one supply would be more sensitive to the other one's satisfaction, while in turn, such constraint is looser for the other supply.

$$\frac{\dot{V}_{f,i}^{t,s}}{\dot{V}_{f,i}^{t,s}} \leq \sum_{i \in \mathcal{N}} \frac{P_{r,i}^{t,s}}{P_i^{t,s}} \varpi_{i,i}^{N,N} \Lambda_{i,i}^{N,N}, \quad \forall i \in \mathcal{N}, t \in T, s \in S \quad (18)$$

$$\frac{P_{r,i}^{t,s}}{P_i^{t,s}} \leq \sum_{i \in \mathcal{N}} \frac{\dot{V}_{f,i}^{t,s}}{\dot{V}_{f,i}^{t,s}} \xi_{i,i}^{N,N} \Lambda_{i,i}^{N,N}, \quad \forall i \in \mathcal{N}, t \in T, s \in S \quad (19)$$

3. Two-stage risk-averse stochastic restoration for CPW system

This section provides formulations for risk-averse stochastic restoration of a CPW system, and describes performance metric (objective), WDS/PDS model and operation, and constraints.

3.1. Performance metric

The performance metric for CPW system is determined as aggregate service level for power and water supply, weighted by their importance, as set by the interruption cost of different consumers [7]. The conceptual curve for CPW system performance affected by disruptions is shown in Fig. 2, which comprises three states: (1) pre-event normal operation state, (2) post-event degraded state, and (3) restored state. Here, we specifically focus on modeling the two states of “start of restoration” and “end of restoration”. During pre-event state, CPW system works in normal operation with full loads supported. When the disruption happens, system performance dramatically drops to the lowest level, R_{pe} . After faults detection and decision-making in the degraded state, restoration starts with multiple measures to promptly improve the system performance from the worst (R_{pe}) to the highest achievable level (R_{pr}), that can be achieved with potential restoration measures as described next.

Here, potential restoration measures include (a) damaged network reconfiguration, (b) pre-allocation of fuels for DGs, (c) energy/water flow dispatch, (d) management of components operation (DG, tank, pump, small PHS) and (e) load curtailments. With further consideration of uncertainties induced by supply and demand conditions, these measures are divided into two stages in stochastic restoration:

First stage: Restoration measures (a) and (b) are to reconnect the blackout power loads with local power sources and determine the allocation of fuel from stock to DGs. This is a one-time decision applied

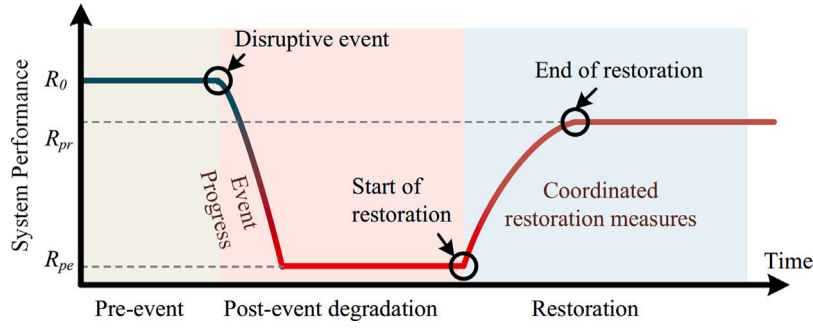


Fig. 2. The conceptual curve for CPW system performance.

at the beginning of restoration state, before addressing uncertainties in power generations of wind/solar renewables and power/water demands.

Second stage: Restoration measures (c), (d) and (e) are to restore service of CPW system during restoration state, after all uncertainties cleared.

3.2. Risk-averse objective

The objective here is to minimize total interruption cost of losing water and power service, and to manage risks. Thus lower objective value implies higher performance. Generally, the algorithm will select loads with higher interruption cost to survive and those with lower cost to sacrifice. To deal with uncertainties and manage risks (wind/solar generations and power/water demands), a risk-averse approach [35] is utilized to optimize composition of Conditional Value at Risk (CVaR) and the expectation of uncertain scenarios, as shown in (20) to (23). For each scenario, the total cost $Cost_s$ is aggregated interruption costs for both power and water loads. v is the mean value at risk (VaR) of solutions for all scenarios and Δ_s is the excessive amount above the VaR. CVaR computes the average of $(1 - \alpha)\%$ worst-case loss scenarios above a thread with weighting of λ .

$$\min \mathbb{E} \{Cost_s\} + CVaR \quad (20)$$

$$Cost_s = \Delta T \sum_{i \in \mathcal{N}} \left[\sum_{t \in \mathcal{T}} w_{w,i} (\mathcal{W}_i^{t,s} - \mathcal{W}_{r,i}^{t,s}) + \sum_{i \in \mathcal{N}} w_{p,i} (P_i^{t,s} - P_{r,i}^{t,s}) \right], \quad \forall s \in \mathcal{S} \quad (21)$$

$$CVaR = \lambda \left(v + \frac{1}{1 - \alpha} \frac{1}{n} \sum_{s=1}^n \Delta_s \right) \quad (22)$$

$$\Delta_s \geq \max \{Cost_s - v, 0\}, \quad \forall s \in \mathcal{S} \quad (23)$$

3.3. Network reconfiguration and fuel pre-allocation

Power network reconfiguration is to adjust PDS topology, pick up disconnected loads and maintain radial operation [36]. Here, the single commodity flow method is applied to determine the status of line linkage. It selects a potential root bus and deploys fictitious flows on the network to guarantee radiality, as formulated in (24) to (29). In the below, Eq. (24) constrains the consistency of the linkage status, (25) represents the conservation of fictitious flow, (26) constrains the value of each fictitious flow according to linkage status, (27) indicates the radiality of the power distribution network, and (28) to (29) indicate that the fictitious demand of non-root nodes in the network is set to 1. Lastly, Eq. (30) indicates fuel pre-allocation for three DGs, which is limited by stock capacity.

$$\alpha_{ij}^P - \alpha_{ji}^P = 0, \quad \forall (i, j) \in \mathcal{E} \quad (24)$$

$$F_{ij} + F_{ji} = 0, \quad \forall (i, j) \in \mathcal{E} \quad (25)$$

$$-M^P \alpha_{ij}^P \leq F_{ij} \leq M^P \alpha_{ij}^P, \quad \forall (i, j) \in \mathcal{E} \quad (26)$$

$$\sum_{(i,j) \in \mathcal{E}} \alpha_{ij}^P = 2 \left(|\mathcal{N}| - \sum_{i \in \mathcal{N}} \beta_i^P \right), \quad \forall i \in \mathcal{N} \quad (27)$$

$$\sum_{(i,j) \in \mathcal{E}} F_{ij} + 1 \geq -M^P \beta_i^P (N_{rt,i} + N_{ft,i}), \quad \forall i \in \mathcal{N} \quad (28)$$

$$\sum_{(i,j) \in \mathcal{E}} F_{ij} + 1 \leq M^P \beta_i^P (N_{rt,i} + N_{ft,i}), \quad \forall i \in \mathcal{N} \quad (29)$$

$$\sum_{m \in \mathcal{M}} E_{ex,m} \leq E_{ex}^{cap}, \quad \forall m \in \mathcal{M} \quad (30)$$

3.4. WDS operation model

WDS delivers potable water to consumers via pipe network. It is represented via an undirected graph: $\mathcal{G} = (\mathcal{N}, \mathcal{E})$. Compared to reported network flow model (e.g. [21,22]), the proposed WDS operation model considers more physical features, e.g. water head loss, node pressure and leakage.

$$\sum_{o \in \mathcal{O}} \mathcal{Q}_{C,o}^{t,s} \Lambda_{o,i}^{0,\mathcal{N}} + \sum_{o' \in \mathcal{O}'} \mathcal{Q}_{f,o'}^{t,s} \Lambda_{o',i}^{0',\mathcal{N}} = \sum_{k \in \mathcal{T}} (\mathcal{Q}_k^{t,s} - \mathcal{D}_k^{t,s}) \Lambda_{k,i}^{\mathcal{T},\mathcal{N}} + \mathcal{W}_{r,i}^{t,s} + \mathcal{W}_{l,i}^{t,s} + \sum_{(i,j) \in \mathcal{E}} \mathcal{Q}_{t,j}^{t,s}, \quad \forall i \in \mathcal{N}, t \in \mathcal{T}, s \in \mathcal{S} \quad (31)$$

$$\mathcal{H}_i^{t,s} - \mathcal{H}_j^{t,s} + \sum_{u \in \mathcal{U}} \mathcal{H}_u^{t,s} \Lambda_{u,(i,j)}^{\mathcal{U},\mathcal{E}} = G \times \text{Sign}(\mathcal{Q}_{i,j}^{t,s}) \times |\mathcal{Q}_{i,j}^{t,s}|^{1.852}, \quad \forall (i, j) \in \mathcal{E}, t \in \mathcal{T}, s \in \mathcal{S} \quad (32)$$

$$\mathcal{P}_i^{t,s} = \mathcal{H}_i^{t,s} - \mathcal{E}_{\mathcal{L}_i}, \quad \forall i \in \mathcal{N}, t \in \mathcal{T}, s \in \mathcal{S} \quad (33)$$

$$\begin{cases} \mathcal{W}_{r,i}^{t,s} \leq \mathcal{W}_i^{t,s} \left(\frac{\mathcal{P}_i^{t,s} - \mathcal{P}_i^{\min}}{\mathcal{P}_i^{\text{des}} - \mathcal{P}_i^{\min}} \right)^{\gamma_i^p}, & \text{IF } \mathcal{P}_i^{\min} \leq \mathcal{P}_i^{t,s} \leq \mathcal{P}_i^{\text{des}} \\ \mathcal{W}_{r,i}^{t,s} = 0, & \text{IF } \mathcal{P}_i^{t,s} \leq \mathcal{P}_i^{\min} \end{cases} \quad (34)$$

$$\forall i \in \mathcal{N}, t \in \mathcal{T}, s \in \mathcal{S} \quad (34)$$

$$\begin{cases} \mathcal{W}_{l,i}^{t,s} = \ell_{c,i} (\mathcal{P}_i^{t,s})^c, & \text{IF } \mathcal{P}_i^{t,s} \geq \mathcal{P}_i^{\min} \\ \mathcal{W}_{l,i}^{t,s} = 0, & \text{Otherwise} \end{cases}, \quad \forall i \in \mathcal{N}, t \in \mathcal{T}, s \in \mathcal{S} \quad (35)$$

$$\Delta UD_i^{t,s} = \mathcal{W}_i^{t,s} - \mathcal{W}_{r,i}^{t,s}, \quad \forall i \in \mathcal{N}, t \in \mathcal{T}, s \in \mathcal{S} \quad (36)$$

$$Buf_i^{t,s} = \sum_{t-\Delta t}^t \Delta UD_i^{t,s}, \quad \forall i \in \mathcal{N}, t \in \mathcal{T}, s \in \mathcal{S} \quad (37)$$

$$0 \leq \mathcal{W}_{r,i}^{t,s} \leq \mathcal{W}_i^{t,s} + c_i^{t,s} Buf_i^{t,s}, \quad \forall i \in \mathcal{N}, t \in \mathcal{T}, s \in \mathcal{S} \quad (38)$$

Eq. (31) computes water flows in each pipe based on mass conservation. The head loss induced by pipe flow are formulated using Hazen-Williams equation, via (32). Pressure-driven constraints prescribe that less, or even no water could be consumed if the pressure dropped below design/minimum pressure for water nodes, as formulated in (33) and (34). Together with water inflow at each node, leakage can exist and is proportional to nodal pressure, which can be modeled as a piece-wise function in (35). Delayed water demand is a specific feature for water consumption, which describes that if the demand at current time step is not fully supplied, but can be supplemented in a near feature, then there is no cost been counted.

Eq. (36) calculates the unmet water demand, and accumulated in a buffer variable in (37). The upper bound of current restoration increases so that to support previous unsatisfied demand, as shown in (38).

3.5. PDS operation model

The model of PDS is developed with another undirected graph, $G = (N, E)$, with connections of loads, backup DGs and rooftop renewables.

$$\begin{aligned} & P_{DG,m}^{t,s} + P_{w,i}^{t,s} + P_{pv,i}^{t,s} - P_{r,i}^{t,s} - \sum_{i \in \mathcal{N}} (P_{P,i}^{t,s} - P_{\mathcal{H},i}^{t,s}) \Lambda_{i,t}^{N,N} \\ &= \sum_{u \in \mathcal{U}} \Gamma_u^{t,s} \Lambda_{u,i}^{N,N} + \sum_{o \in \mathcal{O}} \Gamma_o^{t,s} \Lambda_{o,i}^{N,N} + \sum_{o' \in \mathcal{O}'} \Gamma_{o'}^{t,s} \Lambda_{o',i}^{N,N} + \sum_{(i,j) \in E} P_{ij}^{t,s} + \sum_{i \in \mathcal{N}} \Gamma_{wb,i}^{t,s} \Lambda_{i,i}^{N,N}, \end{aligned} \quad (39)$$

$$Q_{DG,m}^{t,s} + Q_{w,i}^{t,s} + Q_{pv,i}^{t,s} - Q_{r,i}^{t,s} = \sum_{(i,j) \in E} Q_{ij}^{t,s},$$

$$\forall m \in \mathbf{M}, i \in \mathbf{N}, t \in \mathbf{T}, s \in \mathbf{S} \quad (40)$$

$$0 \leq P_{DG,m}^{t,s} \leq P_{DG,m}^{\max}, \forall m \in \mathbf{M}, t \in \mathbf{T}, s \in \mathbf{S} \quad (41)$$

$$-Q_{DG,m}^{\max} \leq Q_{DG,m}^{t,s} \leq Q_{DG,m}^{\max}, \forall m \in \mathbf{M}, t \in \mathbf{T}, s \in \mathbf{S} \quad (42)$$

$$E_{DG,m}^{t,s} = E_{DG,m}^{t-1,s} - P_{DG,m}^{t,s} \Delta T + E_{ex,m} \Xi_{m,t}^E, \quad (43)$$

$$\forall m \in \mathbf{M}, t \in \mathbf{T}, s \in \mathbf{S} \quad (44)$$

$$\begin{aligned} & 0 \leq E_{DG,m}^{t,s} \leq E_{DG,m}^{\max}, \forall m \in \mathbf{M}, t \in \mathbf{T}, s \in \mathbf{S} \\ & -M^P (1 - \alpha_{ij}^P) \leq V_i^{t,s} - V_j^{t,s} - \frac{r_{ij} P_{ij}^{t,s} + x_{ij} Q_{ij}^{t,s}}{V_0} \end{aligned} \quad (45)$$

$$\leq M^P (1 - \alpha_{ij}^P), \quad \forall (i, j) \in E, t \in \mathbf{T}, s \in \mathbf{S} \quad (46)$$

$$-\alpha_{ij}^P S_{ij}^{\max} \leq P_{ij}^{t,s}, Q_{ij}^{t,s} \leq \alpha_{ij}^P S_{ij}^{\max}, \quad (47)$$

$$\forall (i, j) \in E, t \in \mathbf{T}, s \in \mathbf{S} \quad (48)$$

$$-\sqrt{2} \alpha_{ij}^P S_{ij}^{\max} \leq P_{ij}^{t,s} + Q_{ij}^{t,s}, P_{ij}^{t,s} - Q_{ij}^{t,s} \leq \sqrt{2} \alpha_{ij}^P S_{ij}^{\max}, \quad (49)$$

$$\forall (i, j) \in E, t \in \mathbf{T}, s \in \mathbf{S} \quad (50)$$

$$0 \leq P_{r,i}^{t,s} \leq P_i^{t,s}, \quad \forall i \in \mathbf{N}, t \in \mathbf{T}, s \in \mathbf{S} \quad (51)$$

$$0 \leq P_{w,i}^{t,s}, P_{pv,i}^{t,s} \leq \overline{P_{w,i}^{t,s}}, \overline{P_{pv,i}^{t,s}}, \quad \forall i \in \mathbf{N}, t \in \mathbf{T}, s \in \mathbf{S} \quad (52)$$

$$Q_{r,i}^{t,s} = P_{r,i}^{t,s} \tan(\cos^{-1} \varphi_{r,i}), \quad \forall i \in \mathbf{N}, t \in \mathbf{T}, s \in \mathbf{S} \quad (53)$$

$$Q_{w,i}^{t,s} = P_{w,i}^{t,s} \tan(\cos^{-1} \varphi_{w,i}), \quad \forall i \in \mathbf{N}, t \in \mathbf{T}, s \in \mathbf{S} \quad (54)$$

$$Q_{pv,i}^{t,s} = P_{pv,i}^{t,s} \tan(\cos^{-1} \varphi_{pv,i}), \quad \forall i \in \mathbf{N}, t \in \mathbf{T}, s \in \mathbf{S} \quad (55)$$

$$V^{\min} \leq V_i^{t,s} \leq V^{\max}, \quad \forall i \in \mathbf{N}, t \in \mathbf{T}, s \in \mathbf{S} \quad (56)$$

As shown in (39) and (40), active/reactive power flows in PDS are modeled with DistFlow and integrating the power conservation from renewables, DGs, as well as WDS facilities, i.e., small PHS, pump, desalination/treatment plants. Backup DG is a set of local generators in normally-off mode and provide emergency backup power during a grid outage [37]. This paper models backup DG as fuel-based generators [36]. Eqs. (41) and (42) formulate the limits of active/reactive generations. Constraints of fuel usage and capacity are indicated in (43) and (44), where the initial state, $E_{DG,m}^0$ is set to be $E_{DG,m}^{\max}$ and complementary fuel arrives when $\Xi_{m,t} = 1$. Ranges for bus voltage and active/reactive/apparent power flow are in (45) to (47). Eq. (48) to (53) are constraints for renewable generation and power restoration.

4. Solution method

4.1. Scenario-based stochastic restoration model

4.1.1. Uncertainty modeling

Rooftop wind and solar energy generations depend on wind speed ($v_w^{t,s}$) and solar irradiation ($H_{pv}^{t,s}$) respectively. Their stochastic variations

or forecasting errors usually follow the Weibull and Beta distribution [38] as in (54) and (55). As for the water and power demands which are mainly related to consumer behavior, normal distribution can be employed to represent the demand forecasting errors [28].

$$f_{v_w^{t,s}} = \frac{k_w}{\lambda_w} \left(\frac{v_w^{t,s}}{\lambda_w} \right)^{k_w-1} \exp \left(-\frac{v_w^{t,s}}{\lambda_w} \right) \quad (54)$$

$$f_{H_{pv}^{t,s}} = \left(H_{pv}^{t,s} \right)^{\theta-1} (1 - H_{pv}^{t,s})^{\psi-1} N_F \quad (55)$$

$$f_{P_i^{t,s}} = 1 / \left(\sqrt{2\pi} \sigma_p \right) \exp \left[-\left(P_i^{t,s} - \mu_p \right)^2 / (2\sigma_p^2) \right] \quad (56)$$

$$f_{W_i^{t,s}} = 1 / \left(\sqrt{2\pi} \sigma_w \right) \exp \left[-\left(W_i^{t,s} - \mu_w \right)^2 / (2\sigma_w^2) \right] \quad (57)$$

4.1.2. Scenario generation and reduction

Based on the probability distributions of diverse uncertain variables from (54) to (57), Latin hypercube sampling method is applied to generate near-random samples of parameter values, i.e. scenarios. To release the computation burden from large number of samples, simultaneous backward reduction method is used to select a smaller but representative set of scenarios for the practical applications [28].

4.2. Linearization and convexification

In the above equations, nonlinearity and non-convexity manifest in (1), (4), (11), (12), (32), (34) and (35), which lead to a nonlinear programming problem. Herein, linearization and convexification strategies are exploited to reduce computation burden.

4.2.1. Symmetric approximation

Eq. (32) describes the water head loss using Hazen–Williams equation, which is a centrosymmetric function. The target of linearization is to eliminate exponent and absolute value sign. Compared with direct linearization in Ref. [33], this work applies a binary variable, β_{ij}^W , to denote the direction of flow, and convert a directed flow, $\mathbb{Q}_{ij}^t$, to an undirected flow, $\mathbb{Q}_{U,ij}^t$, as in (58) to (62). The following linearization (63) is mainly referenced from [33]. The difference is that the term to be linearized is $\mathbb{Q}_{U,ij}^t$, instead of $\mathbb{Q}_{ij}^t$, which introduces one binary variable for the direction, but reduces binary variables for piece-wise linearization by half. Note that Eqs. (58) to (62) hold for all $(i, j) \in \mathcal{E}$, $t \in \mathbf{T}$, and $s \in \mathbf{S}$ (see Fig. 3).

$$\beta_{ij}^W + \beta_{ji}^W = 1 \quad (58)$$

$$-M^W \beta_{ij}^W \leq \mathbb{Q}_{ij}^{t,s} - \mathbb{Q}_{U,ij}^{t,s} \leq M^W \beta_{ij}^W \quad (59)$$

$$-M^W (1 - \beta_{ij}^W) \leq \mathbb{Q}_{ij}^{t,s} + \mathbb{Q}_{U,ij}^{t,s} \leq M^W (1 - \beta_{ij}^W) \quad (60)$$

$$-M^W (1 - \beta_{ij}^W) \leq \mathcal{H}_i^{t,s} - \mathcal{H}_j^{t,s} - \mathcal{L}_{ij}^{t,s} \leq M^W (1 - \beta_{ij}^W) \quad (61)$$

$$-M^W \beta_{ij}^W \leq \mathcal{H}_i^{t,s} - \mathcal{H}_j^{t,s} + \mathcal{L}_{ij}^{t,s} \leq M^W \beta_{ij}^W \quad (62)$$

$$\begin{cases} \mathbb{Q}_{U,ij}^{t,s} = \sum_{p \in \mathcal{P}} q_p \lambda_{i,j,p}^{t,s}, & \mathcal{L}_{ij}^{t,s} = \sum_{p \in \mathcal{P}} \ell_p \lambda_{i,j,p}^{t,s} \\ \sum_{p \in \mathcal{P}} \gamma_{i,j,p}^{t,s} = 1, & \sum_{p \in \mathcal{P}} \lambda_{i,j,p}^{t,s} = 1 \\ \lambda_{i,j,p}^{t,s} \leq \gamma_{i,j,p-1}^{t,s} + \gamma_{i,j,p}^{t,s}, & \forall p \in \mathcal{P} \setminus \{1, \mathcal{P}\} \\ \lambda_{i,j,1}^{t,s} \leq \gamma_{i,j,1}^{t,s}, & \lambda_{i,j,\mathcal{P}}^{t,s} \leq \gamma_{i,j,\mathcal{P}-1}^{t,s}, \quad 0 \leq \lambda_{i,j,p}^{t,s} \leq 1 \end{cases} \quad (63)$$

4.2.2. Big-M method

Eqs. (34) and (35) indicate the water supply and leakage varying with different pressure conditions. Considering their piece-wise linear property, big M method is used to help determine the pressure interval. To achieve this, two sets of binary variables are declared: $\mathcal{K}_{ck1,i}^{t,s}$ and $\mathcal{K}_{ck2,i}^{t,s}$. The Eqs. (64) to (69) hold for all $i \in \mathcal{N}$, $t \in \mathbf{T}$, and $s \in \mathbf{S}$.

$$-M^W (1 - \mathcal{K}_{ck1,i}^{t,s}) \leq \mathcal{P}_i^{t,s} - \mathcal{P}_i^{des} \leq M^W \mathcal{K}_{ck1,i}^{t,s} \quad (64)$$

$$-M^W (1 - \mathcal{K}_{ck2,i}^{t,s}) \leq \mathcal{P}_i^{t,s} - \mathcal{P}_i^{\min} \leq M^W \mathcal{K}_{ck2,i}^{t,s} \quad (65)$$

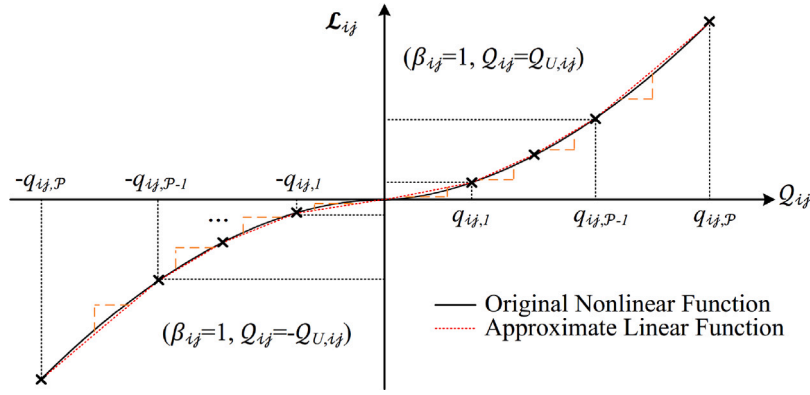


Fig. 3. Symmetric approximation for Hazen-Williams equation.

$$\mathcal{W}_{r,i}^{t,s} - \mathcal{W}_i^{t,s} \frac{\mathcal{P}_i^{t,s} - \mathcal{P}_i^{\min}}{\mathcal{P}_i^{\text{des}} - \mathcal{P}_i^{\min}} \leq M^{\mathcal{W}} \mathcal{K}_{ck1,i}^{t,s} \quad (66)$$

$$\mathcal{W}_{r,i}^{t,s} \leq M^{\mathcal{W}} \mathcal{K}_{ck2,i}^{t,s} \quad (67)$$

$$-M^{\mathcal{W}} (1 - \mathcal{K}_{ck1,i}^{t,s}) \leq \mathcal{W}_{l,i}^{t,s} - \ell_{e,i} (\mathcal{P}_i^{t,s})^{\gamma_i^e} \leq M^{\mathcal{W}} (1 - \mathcal{K}_{ck1,i}^{t,s}) \quad (68)$$

$$-M^{\mathcal{W}} \mathcal{K}_{ck2,i}^{t,s} \leq \ell_{e,i} \leq M^{\mathcal{W}} \mathcal{K}_{ck2,i}^{t,s} \quad (69)$$

4.2.3. McCormick envelopes

Eqs. (4) and (5) describe the operation of pumped-hydro storage, which contains the productions of two variables. For the computation efficiency, the McCormick envelop method is used to relax and eliminate the nonlinearities, as shown in (70) and (71).

$$\begin{cases} \Pi_{\mathcal{G},\mathcal{F}_k}^{t,s} = \mathcal{H}_i^{t,s} \mathcal{G}_{\mathcal{F}_k}^{t,s} \\ \eta^p P_{P,i}^{t,s} = \rho g \mathcal{H}_{r,i} \mathcal{G}_{\mathcal{F}_k}^{t,s} - \rho g \Pi_{\mathcal{G},\mathcal{F}_k}^{t,s} \\ \Pi_{\mathcal{G},\mathcal{F}_k}^{t,s} \geq \mathcal{H}_i^{\min} \mathcal{G}_{\mathcal{F}_k}^{t,s} + H_i^{t,s} \mathcal{G}_{\mathcal{F}_k}^{\min} - H_i^{\min} \mathcal{G}_{\mathcal{F}_k}^{\min} \\ \Pi_{\mathcal{G},\mathcal{F}_k}^{t,s} \geq \mathcal{H}_i^{\max} \mathcal{G}_{\mathcal{F}_k}^{t,s} + H_i^{t,s} \mathcal{G}_{\mathcal{F}_k}^{\max} - H_i^{\max} \mathcal{G}_{\mathcal{F}_k}^{\max} \\ \Pi_{\mathcal{G},\mathcal{F}_k}^{t,s} \leq \mathcal{H}_i^{\max} \mathcal{G}_{\mathcal{F}_k}^{t,s} + H_i^{t,s} \mathcal{G}_{\mathcal{F}_k}^{\min} - H_i^{\max} \mathcal{G}_{\mathcal{F}_k}^{\min} \\ \Pi_{\mathcal{G},\mathcal{F}_k}^{t,s} \leq \mathcal{H}_i^{\min} \mathcal{G}_{\mathcal{F}_k}^{t,s} + H_i^{t,s} \mathcal{G}_{\mathcal{F}_k}^{\max} - H_i^{\min} \mathcal{G}_{\mathcal{F}_k}^{\max} \end{cases} \quad (70)$$

$$\begin{cases} \Pi_{\mathcal{D},\mathcal{F}_k}^{t,s} = \mathcal{H}_i^{t,s} \mathcal{D}_{\mathcal{F}_k}^{t,s} \\ P_{P,i}^{t,s} = \eta^p \rho g \mathcal{H}_{r,i} \mathcal{D}_{\mathcal{F}_k}^{t,s} - \eta^p \rho g \Pi_{\mathcal{D},\mathcal{F}_k}^{t,s} \\ \Pi_{\mathcal{D},\mathcal{F}_k}^{t,s} \geq \mathcal{H}_i^{\min} \mathcal{D}_{\mathcal{F}_k}^{t,s} + H_i^{t,s} \mathcal{D}_{\mathcal{F}_k}^{\min} - H_i^{\min} \mathcal{D}_{\mathcal{F}_k}^{\min} \\ \Pi_{\mathcal{D},\mathcal{F}_k}^{t,s} \geq \mathcal{H}_i^{\max} \mathcal{D}_{\mathcal{F}_k}^{t,s} + H_i^{t,s} \mathcal{D}_{\mathcal{F}_k}^{\max} - H_i^{\max} \mathcal{D}_{\mathcal{F}_k}^{\max} \\ \Pi_{\mathcal{D},\mathcal{F}_k}^{t,s} \leq \mathcal{H}_i^{\max} \mathcal{D}_{\mathcal{F}_k}^{t,s} + H_i^{t,s} \mathcal{D}_{\mathcal{F}_k}^{\min} - H_i^{\max} \mathcal{D}_{\mathcal{F}_k}^{\min} \\ \Pi_{\mathcal{D},\mathcal{F}_k}^{t,s} \leq \mathcal{H}_i^{\min} \mathcal{D}_{\mathcal{F}_k}^{t,s} + H_i^{t,s} \mathcal{D}_{\mathcal{F}_k}^{\max} - H_i^{\min} \mathcal{D}_{\mathcal{F}_k}^{\max} \end{cases} \quad (71)$$

4.2.4. Convex relaxation

Constraint (12) is a quadratic equation which is nonconvex. It can be relaxed as shown in (72). The rationale is referenced from [29].

$$\begin{cases} \eta_u \Gamma_u^{t,s} \geq f_{u,c} \sum_{(i,j) \in \mathcal{E}} (f_{u,a} \mathcal{Q}_{i,j}^{t,s^2} + f_{u,b} \mathcal{Q}_{i,j}^{t,s}) \Lambda_{u,i,j}^{\mathcal{U},\mathcal{E}} \\ \eta_u \Gamma_u^{t,s} \leq f_{u,c} \sum_{(i,j) \in \mathcal{E}} (f_{u,a} \mathcal{Q}_{i,j}^{\max} + f_{u,b}) \Lambda_{u,i,j}^{\mathcal{U},\mathcal{E}} \end{cases}, \forall u \in \mathcal{U}, t \in T, s \in S \quad (72)$$

4.2.5. Piece-wise linearization

Eq. (1) formulates the energy consumption of desalination plant. The fraction and quadratic terms are linearized by piece-wise approach [32].

$$\begin{cases} \Gamma_o^{t,s} = e_o^{t,s} \mathcal{Q}_{C,o}^{t,s} \\ e_o^{t,s} = e_{o,1}^{t,s} + e_{o,2}^{t,s} + e_{o,3}^{t,s} + e_{o,4}^{t,s} \\ \mathcal{Q}_{C,o}^{t,s} = \mathcal{Q}_{C,o,1}^{t,s} + \mathcal{Q}_{C,o,2}^{t,s} + \mathcal{Q}_{C,o,3}^{t,s} + \mathcal{Q}_{C,o,4}^{t,s} \\ 0 \leq \mathcal{Q}_{C,o,1}^{t,s} \leq 0.25 \mathcal{Q}_{C,o}^{\max} z_{o,1}^{t,s} \\ 0.25 z_{o,2}^{t,s} \mathcal{Q}_{C,o}^{\max} \leq \mathcal{Q}_{C,o,2}^{t,s} \leq 0.5 \mathcal{Q}_{C,o}^{\max} z_{o,2}^{t,s} \\ 0.5 z_{o,3}^{t,s} \mathcal{Q}_{C,o}^{\max} \leq \mathcal{Q}_{C,o,3}^{t,s} \leq 0.75 \mathcal{Q}_{C,o}^{\max} z_{o,3}^{t,s} \\ 0.75 z_{o,4}^{t,s} \mathcal{Q}_{C,o}^{\max} \leq \mathcal{Q}_{C,o,4}^{t,s} \leq \mathcal{Q}_{C,o}^{\max} z_{o,4}^{t,s} \end{cases} \quad (73)$$

After the linearization and convexification, the original nonlinear programming problem becomes MISOCP, which can be better solved by off-the-shelf solvers.

5. Case study and numerical results

5.1. Case study

A CPW system comprising a 25-node water network [39] and a 33-bus power [36] is applied to verify the effectiveness of proposed framework. The illustration of PDS and WDS as well as interdependencies are in Fig. 4. In PDS, three fuel-based backup DGs are connected with power network, while two desalination and one treatment plant, three tanks (two of them equipped with small PHS), and two pumps are installed in WDS. The installation and base capacity of rooftop renewables are shown in Table 1, and other interdependency connections are shown in Table 2. The disaster event considered has three faulted branches (1,2), (6,26) and (32,33) [36]; while at water side, a main conjunction besides node 21, a pump station at link (2,3), and four pipelines, (5,6), (7,19), (12,22) and (13,24) are disabled.

The time interval is 1 h while total horizon considered is 24 h. Fig. 5(a) and (b) show the forecast generation patterns of wind/solar energy based on wind speed and solar irradiation [28]. Loads are varying by time with forecast profiles applied to every power bus and water node, which are shown in Fig. 5(c) and (d). The daily average interruption costs for power and water load are represented in Fig. 6(a) and (b) [36]. Fig. 6(c) shows a temporal multiplier representing the varying importance of the loads [40], which works as a temporal multiplier on the daily average interruption cost at each bus/node. Hence, the total interruption cost is obtained by summing over all the buses and nodes, as representing the interruption cost from the consumer's viewpoint. For power buses, $V_0/V^{\max}/V^{\min}$ are 1.0/1.05/0.95 p.u. respectively. The capacity of stock fuels is 10MWh, and allocated fuels are assumed to arrive at each DG at $t = 4\text{hr}$. For WDS, $\mathcal{P}_i^{\text{des}}$ and $\mathcal{P}_i^{\min}$ are 10 and 0 meter. $\ell_{e,i}$ is $3.6e-3 \text{ m}^2/\text{h}$ and $c_i = \gamma_i^p = \gamma_i^e = 1$ are set to all water nodes. Other parameters are shown in Table 3 and 4.

In terms of risk management, λ and α are set as 1.0 and 0.8, respectively [35], which means the weight for risk measure is 1, with the worst 20% scenarios considered. It should be noted that all the above input information is available from numerous data resources [36, 39]. Other physical representative data can also be applied without impairing the method's effectiveness. The optimization is modeled by GAMS and solved by Cplex, on a PC with 3.20 GHz 64-bit CPU and 32G RAM. The convergence criteria for the simulation is set at 0.1% change in the objective (Eq. (20)). The computation time for the first and second stage optimization are 456.72 and 5.55 s, respectively.

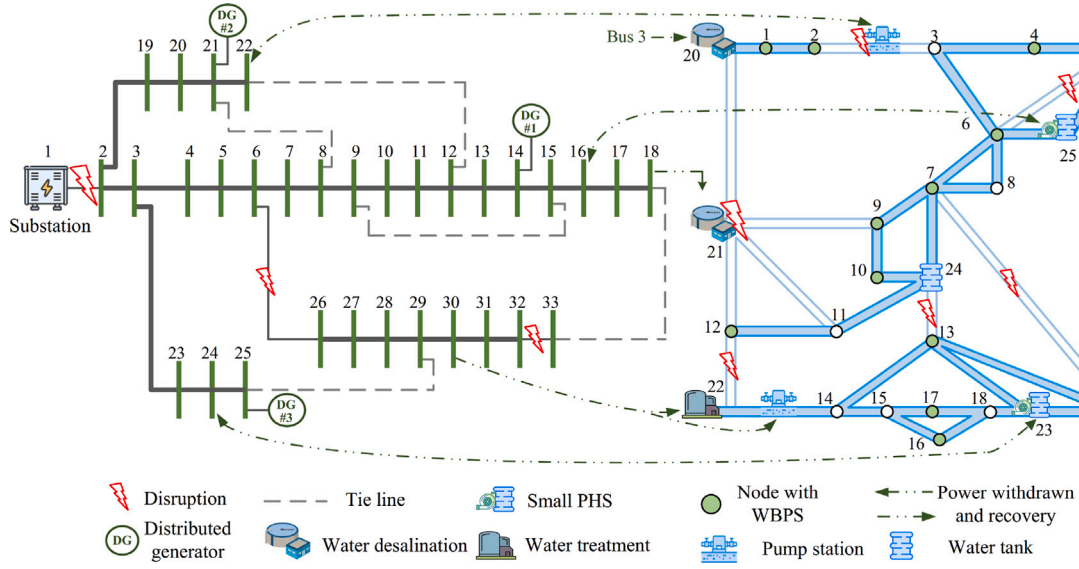


Fig. 4. CPW test system with modified 33-bus power and 25-node water networks.

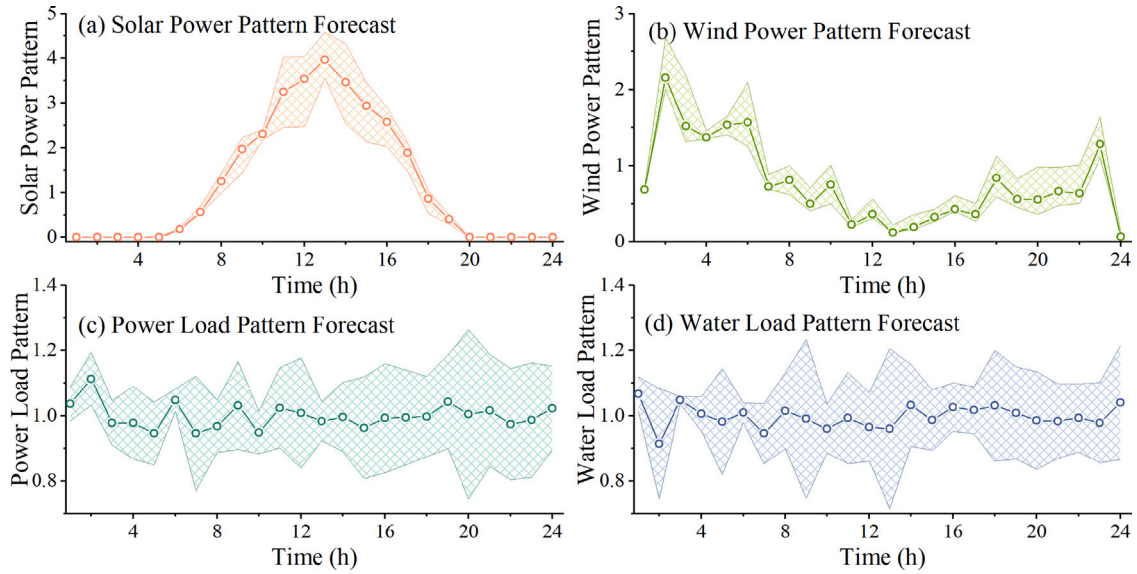


Fig. 5. Forecast for the pattern of (a) solar power, (b) wind power, (c) power demand and (d) water demand. The middle line refers to mean and the bounded area spans 5 to 95 percentile.

Table 1
Rooftop renewables.

Type	Location (Bus #)	Base capacity(kW)
Wind	19, 32	80
Solar	3, 30, 33/8, 17/22, 26	100/80/40

Table 2
Interdependencies.

Type	Connection (Node # - Bus #)
WBPS	1-3, 2-2, 4-5, 5-6, 6-7, 7-8, 9-8, 10-9, 12-11, 13-12, 16-13, 17-14, 19-15
Nexus Load	4-5, 6-7, 9-8, 12-11, 16-13, 17-14

Table 3
Parameters for backup DGs and small PHS.

Backup DGs			Small PHS (Water tanks)			
m/i	$P_{DG,m}^{\max}$	$E_{DG,m}^{\max}$	k/i	η_i^p / η_i^w	$\mathcal{H}_{r,i}^t$	$\eta_i^0 / \eta_i^{\max}$
1/14	1.35	5.8	1/23	0.75/0.7	80	600/1200
2/21	1.35	10	2/24	0.75/—	75	500/1200
3/25	1.35	5.8	3/25	0.75/0.7	75	600/1200

Table 4
Other parameters setting [29,31].

c_u	η_i	p_{pr}	p_a	η_T	$\eta_{f,i}$	$\varpi_{i,i}^{N,N}$	$\xi_{i,i}^{N,N}$	r_{σ}^f
600	0.7	2	0.75	200	195	1.1	1.5	1.5e-3

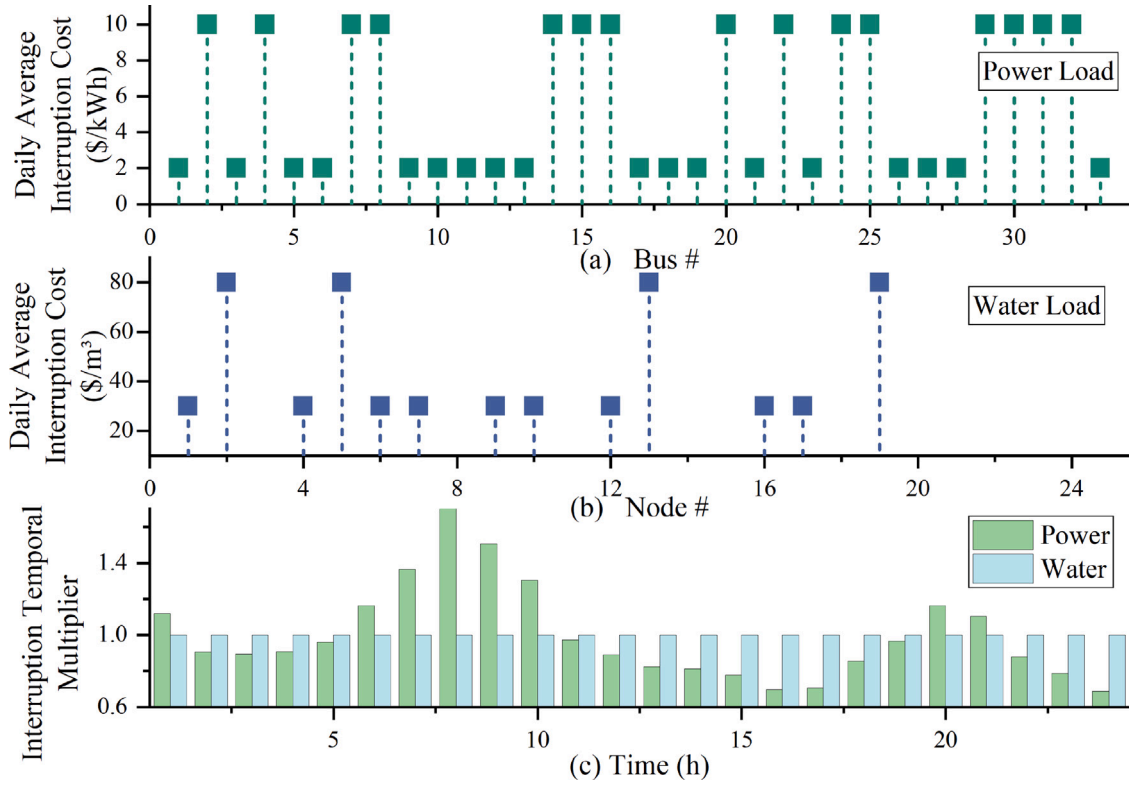


Fig. 6. Interruption cost for (a) power load and (b) water load as well as (c) temporal multiplier which defines varying interruption cost by time.

5.2. Numerical results

In this work, 2000 random scenarios are initially generated with Latin Hypercube Sampling (LHS), based on wind/solar generation and power/water demand predictions in Fig. 5. Then they are reduced to 15 representative ones with corresponding probabilities by Simultaneous Backward Reduction technique [28]. After the simulation, the expectation of total interruption cost is \$323,183 and the risk measure (CVaR) is \$317,993. The decision of power network reconfiguration is shown in Fig. 7. The allocation of fuels is 7.718MWh for DG #2 (bus 21 in island #2) and 2.281MWh for DG #3 (bus 25 in island #3). These decisions refer to restoration measure (a) and (b) of the first stage in Section 3 (A).

As shown in Fig. 7, disconnected loads are picked up via tie lines and switches. After reconfiguration, three radial islands are formed to optimally dispatch energy. The decision considers overall generation capacity and demand of both PDS and WDS, their uncertainties, and delivery losses. Each island is supplied by backup DGs and distributed renewables included within the island. These energies are selectively dispatched to loads according to their interruption cost. Water related components are distributed on three islands, where, desalination plant at node 20, and small PHS at node 23 are connected with island #2; water treatment plant and a neighbor pump station are supported by island #3; and island #1 connects to another small PHS at node 25. With consideration of continuity for power/water supply, fuel-fired DGs are equally arranged on each island.

For the detailed CPW system operation (restoration measures c, d, and e), another 100 scenarios are randomly sampled to run for the reconfigured system. In the rest of this paragraph, scenario #1 is further illustrated for second stage restoration methods. Fig. 8 illustrates the power generation (DGs and renewables) and consumption (loads) and Fig. 9 shows water production (desalination and treatment plants) and usage (loads) for total WDS. The water load curtailment and delayed water demand for each node are visualized in Fig. 10(a) and (b),

while Fig. 10(c) shows power load curtailment for each bus. From the power/water dispatch results, we can observe that:

- (1) All buses are reconnected by reconfiguration (Fig. 7) and partially restored by local energy resources (Fig. 8).
- (2) On each island, the dispatching of DG power is organized in collaboration with renewables. Hence, both renewable potential and fuels are fully utilized.
- (3) The burdens and benefits from WDS are distributed to three islands. For island #2, water-related components result in power consumption for desalination and recovery from small PHS. On island #3, water treatment plant and pump station consume large amount of power. Water components linked with island #1 contains a small PHS and a few WBPS units, with the accumulated influence shows a minor power demand to PDS.
- (4) Small PHSs work as power and water storage facility and the charging/discharging behaviors induce extraction/release of both water and power.
- (5) Loads are selectively curtailed in PDS and WDS based on the interruption cost (importance), as in Figs. 6 and 10.
- (6) Delayed water demand are met on node 5, 6, 7, 9 and 10, etc., which introduces more flexibility to water distribution, and indirectly allows for flexibility in power flow organization.

In summary, the proposed restoration method could coordinate power/water production and system operation, to reduce the total interruption cost and improve the resilience. For this scenario #1, the interruption cost is \$205,776 for power side and \$128,256 for water side.

5.3. Method comparison

To verify the advantages of the proposed method, four commonly used restoration methods are compared below:

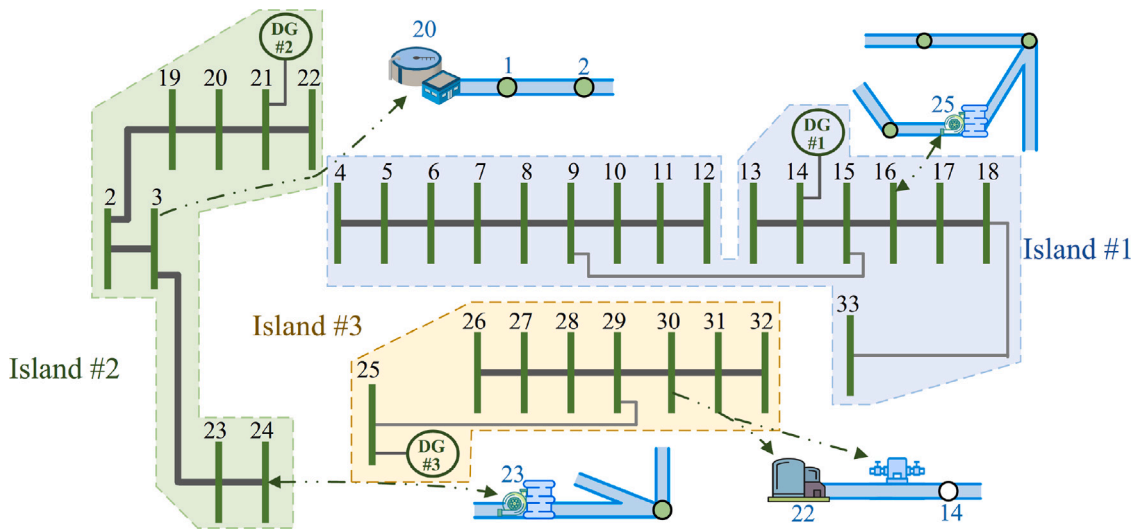


Fig. 7. Power distribution network reconfiguration.

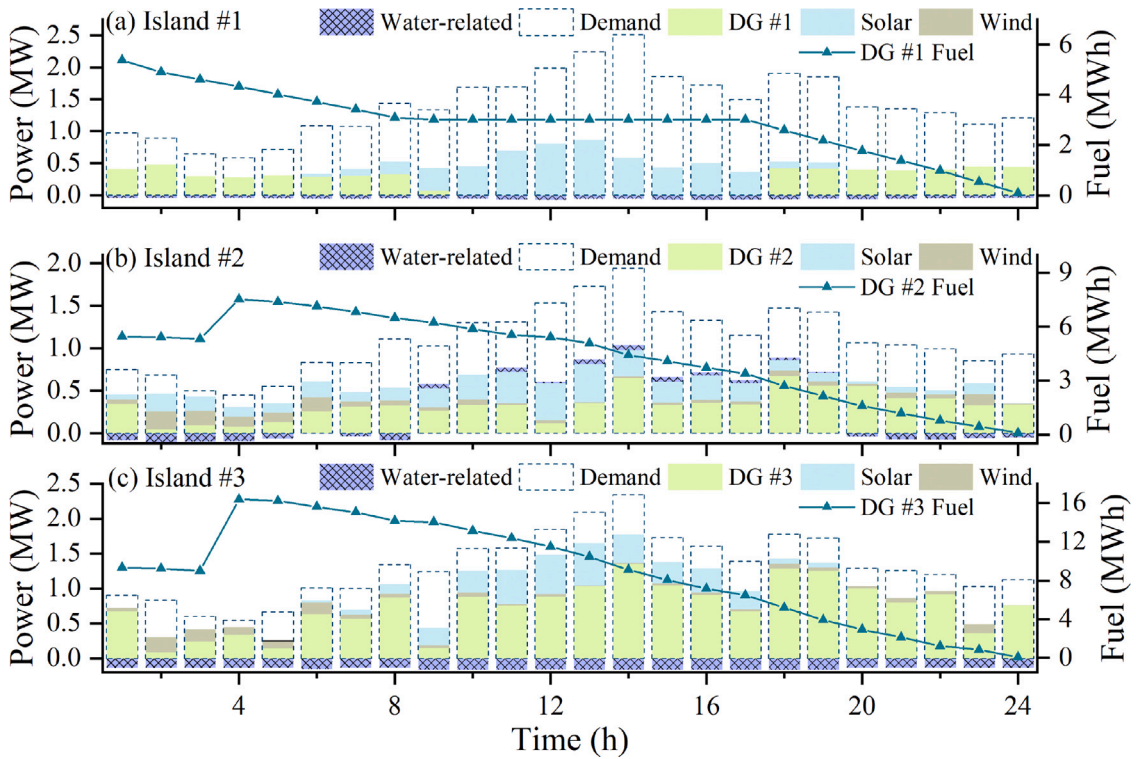


Fig. 8. Solution details for power generation/consumption and DG operation in PDS. Right vertical axis and line chart show the fuel change for DG.

Method #1: No network reconfiguration and fuel pre-allocation. In this method, the network reconfiguration and fuel pre-allocation are ignored. The topology of the power distribution system remains at the disrupted state with all tie lines disconnected. The fuel reserve is set as 0, which indicates the unavailability of allocation.

Method #2: Independently restoring WDS and PDS. In this case, the restorations for PDS and WDS are separately conducted, without considerations of coordination and interdependencies. To represent this, all the decision variables of interdependent linkages are disabled. The state information of the other system is conservatively assumed to be shown by its initial state [41].

Method #3: Restoring the CPW system without dispatching small PHS and rooftop renewables. In this case, the power generated from

renewables and recovered from small PHS is considered as the supplement of the connected bus. To represent this, the upper limit of renewable generation is further constrained by the power demand of connected buses.

Method #4: Restoring the CPW system with the deterministic model. In this case, the uncertainties in generations as well as demands are neglected. Such parameters are then set as their mean values.

Method #5: Restoration with risk-neutral stochastic optimization, i.e., no risk is considered in this setting. To represent this, the term for risk measure is removed from the objective function.

The corresponding interruption costs for 100 sampled scenarios are summarized in Table 5.

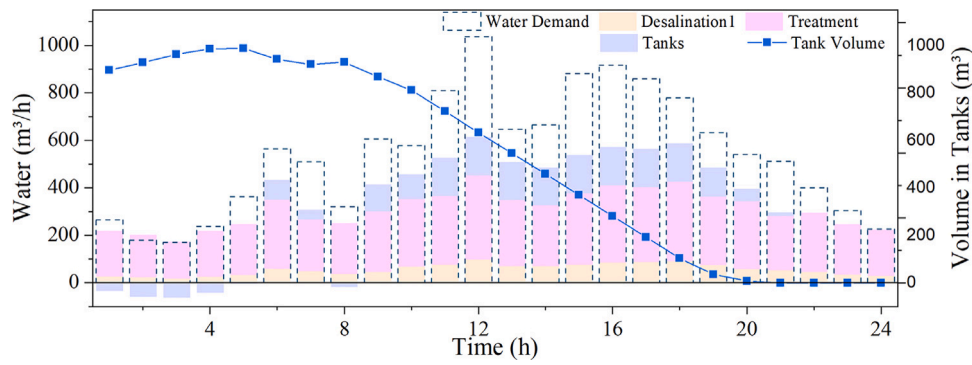


Fig. 9. Solution details for water production/consumption and tanks operation in WDS. Right vertical axis and line chart show the water volume for tanks.

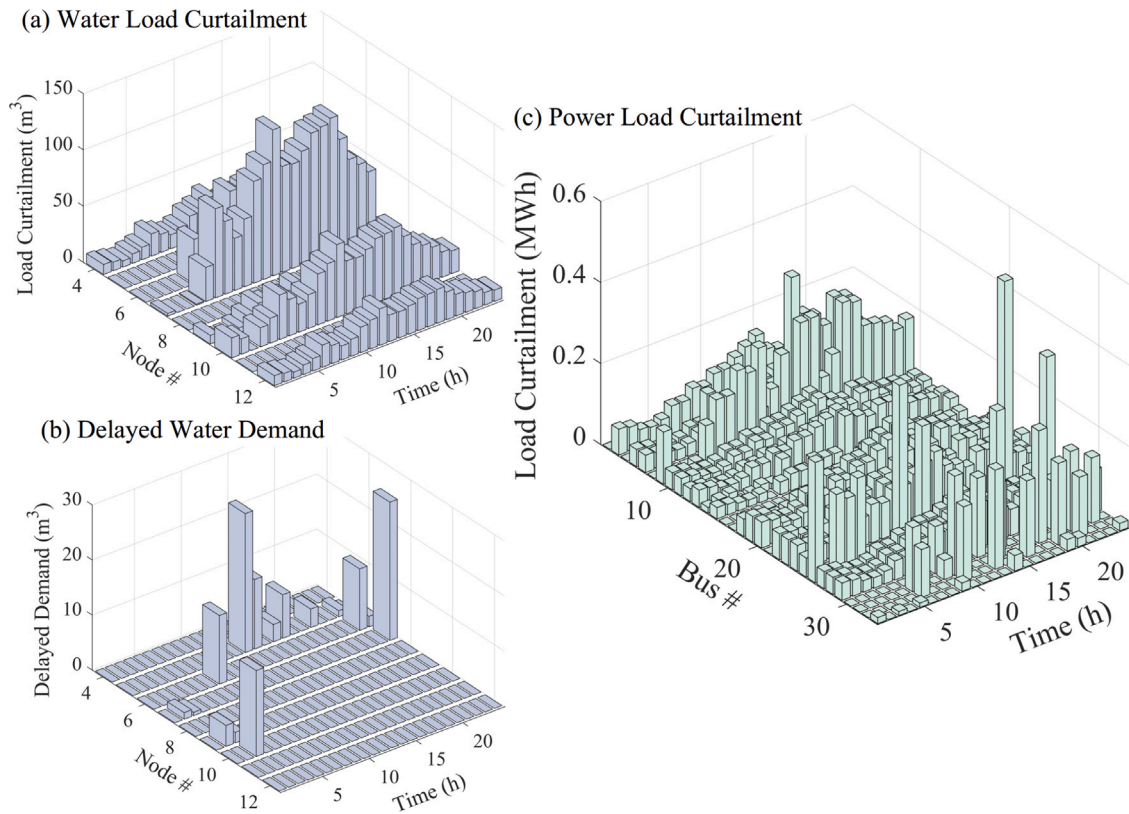


Fig. 10. Results for (a) water load curtailments, (b) delayed water demands and (c) power load curtailments.

Table 5
Simulation results for method #1-5 (\$).

		Method #1	Method #2	Method #3	Method #4	Method #5	Proposed method
First Stage	Cost	591,785	551,709	371,866	330,825	320,670	323,183
	CVaR	601,765	561,209	412,046	—	—	317,993
Second Stage	Average Cost	659,897	567,357	418,428	338,539	331,302	331,932

As seen in Table 5, the cost of **Method #1** is the highest and 83.1% higher on average when compared to the proposed method. This comparison illustrates the importance of network reconfiguration and fuel pre-allocation. The cost of **Method #2** is 70.9% higher on average when compared to the proposed method. This indicates the necessity and efficacy of the coordination between power and water restoration; For **Method #3**, the average cost is 26.1% higher than the proposed method. It is because the power from rooftop renewables is not dispatched through the network and excess energy is wasted. For **Method #4**, the results show 1.99% higher on average than the

proposed method. This comparison supports the consideration of uncertainties in this work. Finally, the comparison with **Method #5** shows that the average cost is close to the proposed method (only 0.2% lower). However, for more extreme scenarios, the costs are 2.57% ~ 6.56% higher than the proposed method (Fig. 11), which adopts risk-averse in all scenarios. This comparison exhibits the necessity of risk consideration for coordinate restoration in the proposed method.

To summarize, it can be seen that the interruption costs are gradually reduced from **Method #1** to #5, with the restoration strategy getting increasingly sophisticated. Moreover, the risk-neutral method

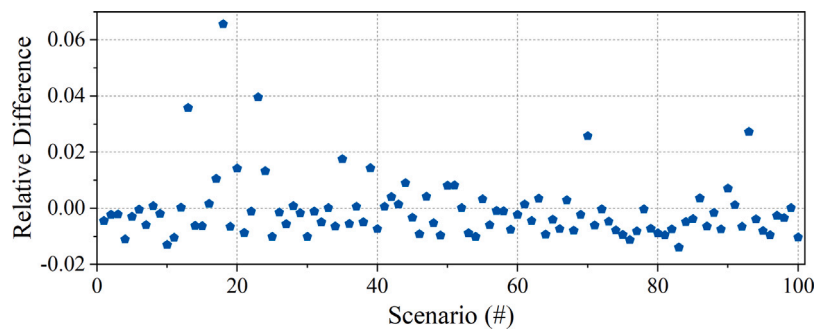


Fig. 11. Relative difference of costs between **Method #5** and proposed method for each scenario, calculated as $\frac{\text{Method \#5} - \text{Proposed Method}}{\text{Proposed Method}}$.

risks poor performance under certain scenarios. These results show that the proposed risk-averse method is effective at lowering CPW interruption costs, mitigating various uncertainties, and flexibly coordinating PDS and WDS restoration.

6. Concluding remarks

In this paper, a coordinated risk-averse restoration method for coupled power and water systems is presented while considering small pumped-hydro storage, rooftop renewables and various uncertainties. Numerical case studies are implemented to verify the effectiveness. It is concluded from the simulation results that:

- (1) The proposed method could effectively coordinate the post-disruption of PDS and WDS to reduce overall interruption cost.
- (2) The proposed method is better than the conventional individual restoration in all sampled scenarios. This shows the necessity of considering the interdependencies within CPW systems.
- (3) The dispatching of small PHS as well as rooftop renewables not only increases possible energy sources but also improves the flexibility of power and water restoration.
- (4) In comparison with the deterministic optimization, the proposed method shows robustness to various uncertainties in the real-time operation, such as solar irradiation, wind speed, power and water demands, etc.
- (5) It is necessary to take into account the risk-averse method to avoid excessive cost under some scenarios.

The developed restoration framework is demonstrated using coupled benchmark systems, but other physical representative data from larger practical systems can be readily integrated into the framework without loss of generality. Regarding the limitations of current work, first, the developed model focuses on the interdependent linkages between PDS and WDS. The individual power and water systems can be modeled in more detail with regard to component behavior. Furthermore, features such as microgrids in power systems, and inertia and time delay characteristics in water transport can be included. The restoration problem studied represents an emergency response problem. Optimizations for pre-disruption preparation, component repair and system recovery can be incorporated as well as the use of robust optimization.

CRediT authorship contribution statement

Yesen Yang: Conceptualization, Methodology, Software, Validation, Writing – original draft. **Zhengmao Li:** Conceptualization, Validation, Writing – review & editing. **Pradeep V. Mandapaka:** Conceptualization, Writing – review & editing. **Edmond Y.M. Lo:** Conceptualization, Methodology, Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data used in this manuscript is from cited references and mentioned/shown in the article.

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